Shaoliang Yang

Curriculum Vitae

Ph.D. Candidate, Mechanical Engineering, Santa Clara University
Expected graduation: June 2027 · Advisor: Prof. Jun Wang
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Research Interests

Generative design; topology optimization; isogeometric analysis; NURBS geometric modeling; LLM agents for design automation; GPU-accelerated FEA/SIMP; architected materials; ML for mechanical engineering.

Education

Ph.D., Mechanical Engineering, Santa Clara University 2023 – present
Advisor: Prof. Jun Wang; Expected June 2027; GPA 3.76/4.00

M.Eng., Control Theory and Control Engineering, Yunnan University 2020 – 2023
Three-year academic (research-track) master's; thesis-based, advisor: Prof. Dongming Zhou. Assessed by WES as equivalent to a U.S. master's degree.

B.E., Automation, Kunming University of Science and Technology 2014 – 2018

Teaching Experience

Grader, MECH 45 Applied Programming for Mechanical Engineers (Sections 1 & 3) Spring 2025
Santa Clara University — assignment feedback and assessment for undergraduate ME programming.

Guest Lecturer (2 hours), Computational Geometry for CAD/CAM
Santa Clara University — lecture connecting geometric representations to CAD/CAM practice.

Research Experience

Research Assistant, Santa Clara University (Prof. Jun Wang) 2023 – present

- Generative/inverse design for curves, surfaces, and 3D architected cellular materials (NURBS-OT, parametric autoencoders, CH-WGAN, SDF reconstruction).
- LLM-driven controllers for SIMP topology optimization (AutoSiMP; adaptive continuation).
- Matrix-free GPU kernels and multigrid solvers for large-scale 3D SIMP systems; NURBS synthesis coupled with IGA.

Graduate Researcher (M.Eng. thesis), Yunnan University (Prof. Dongming Zhou) 2020 – 2023

- Three-year research-track master's devoted to thesis research rather than coursework: deep generative models for low-light image enhancement and denoising.
- First author on three journal papers from this work — *IEEE Trans. Computational Imaging*, *IEEE Signal Processing Letters*, and *Int. J. Machine Learning and Cybernetics*.

Publications

Selected; full list on Google Scholar. Bold = Shaoliang Yang.

Design / geometry / topology optimization

1. **S. Yang**, J. Wang. Style-IH-GAN: Diverse Inverse Homogenization and Continuous Graded Synthesis of Cellular Metamaterials. *Engineering with Computers*, 42:148, 2026. doi:10.1007/s00366-026-02393-3 (open access).

2. **S. Yang**, J. Wang. CH-WGAN: Uncertainty-Aware Generative Design of Three-Dimensional Aperiodic Architected Cellular Materials via Signed Distance Fields. *J. Mechanical Design*, 148(12):121701, 2026. doi:10.1115/1.4071593.
3. **S. Yang**, J. Wang. Interpretable neural basis approximation for periodic cellular surfaces through analytic primitive mixtures. *Computer-Aided Design*, 196:104064, 2026.
4. **S. Yang**, J. Wang, Y. Wang. Large Language Models as Optimization Controllers: Adaptive Continuation for SIMP Topology Optimization. *Advances in Engineering Software*, 223:104304, 2026. doi:10.1016/j.advengsoft.2026.104304. Code: BSD 3-Clause.
5. **S. Yang**, J. Wang. Triple-parametric autoencoder for 2D reparameterization via Bézier, B-spline, and NURBS representations. *Computer-Aided Design*, 189:103936, 2025.
6. **S. Yang**, J. Wang, K. Wang. NURBS-OT: An Advanced Model for Generative Curve Modeling. *J. Mechanical Design*, 147(3):031703, 2025. doi:10.1115/1.4066549.
7. **S. Yang**, J. Wang. Inverse Inference of Aperiodic Structures: Decoding Signed Distance Fields into Learnable Parameters for 3D Reconstruction. ASME IDETC/CIE, 2025. Paper No. DETC2025-168575.
8. **S. Yang**, J. Wang, Y. Wang. NeuralTO-Bench: A Reproducibility Framework and Benchmark Foundation for Classical and Neural Mesh-Free Topology Optimization. ASME IDETC/CIE, 2026. Paper No. IDETC2026-194238 (accepted; not yet online).
9. **S. Yang**, K. Wang, J. Wang. Enhancing Isogeometric Analysis with NURBS-Based Synthesis. ASME IDETC/CIE, 2024. Paper No. DETC2024-142195.
10. Y. Wang, Z. Yashengjiang, B. Yu, **S. Yang**, J. Wang. Semantic Inverse Design of Cellular Structures in Implicit Field Space. ASME IDETC/CIE, 2026. Paper No. IDETC2026-194262 (accepted).

Preprints / under review

11. **S. Yang**, J. Wang. Does Super-Resolution Preserve Defect Evidence? A Low-False-Call Benchmark for Semiconductor Inspection. arXiv:2607.17401, 2026. Under review, *J. Micro/Nanopatterning, Materials, and Metrology* (JM3).
12. **S. Yang**, J. Wang. SafeRestore: Detector-Relative Risk Certificates for Selective Industrial Image Restoration. arXiv:2609.03475, 2026. Preprint.
13. **S. Yang**, J. Wang, Y. Wang. AutoSiMP: Autonomous Topology Optimization from Natural Language via LLM-Driven Problem Configuration and Adaptive Solver Control. arXiv:2603.27000, 2026. Preprint.
14. **S. Yang**, J. Wang, Y. Wang. Matrix-Free 3D SIMP Topology Optimization with Fused Gather-GEMM-Scatter Kernels. arXiv:2604.18020, 2026. Preprint. Code: BSD 3-Clause.
15. **S. Yang**, J. Wang, Y. Wang. A Matrix-Free Galerkin Multigrid Solver and Failure-Mode Screen for Single-GPU 3D SIMP Linear Systems. arXiv:2604.26441, 2026. Code: BSD 3-Clause.
16. **S. Yang**, J. Wang, Y. Wang. When a Positive SIMP Density Floor Is Not Enough: Solver Admissibility and Guarded Floor Selection in Matrix-Free 3D Topology Optimization. arXiv:2607.26382, 2026. Under review, *Comput. Methods Appl. Mech. Eng.* (CMAME). Code: BSD 3-Clause.
17. **S. Yang**, J. Wang, Y. Wang. IterSIMP-σ: Evaluating LLM-Assisted Spatial Interventions in Stress-Aware Topology Optimization. arXiv:2605.19110, 2026. Under revision, *J. Mechanical Design*.
18. **S. Yang**, H. Chu, Z. Yashengjiang, J. Wang. A Property-Registry Contract for Retrieve-or-Refuse Thermal-Mechanical Lattice Search. arXiv:2609.14741, 2026. Code: MIT.
19. **S. Yang**, J. Wang, Y. Wang. Empirical Calibration and Conditional-Reliability Diagnostics for Bearing RUL Prediction under Operating-Regime Shift. arXiv:2607.08273, 2026. Under review, *Quality and Reliability Engineering International*. Code: MIT.
20. **S. Yang**, J. Wang. Large Language Models Can Acknowledge an Impossible Engineering Problem yet Still Report It as Solved. arXiv:2610.06668, 2026. Preprint. Submitted to *Advanced Science*.
21. **S. Yang**, J. Wang. How the Classical Baseline Shapes Machine-Learning Speed Claims in Topology Optimization. arXiv:2610.16051, 2026. Submitted to *Artificial Intelligence Review*.
22. **S. Yang**, J. Wang. Consequence-Guided Clarification: Releasing Minimum-Compliance Topology-Optimized Designs from Incomplete Specifications. arXiv:2610.34957, 2026. Preprint.

Prior learning-based imaging (selected)

23. **S. Yang**, D. Zhou, J. Cao, Y. Guo. LightingNet: An Integrated Learning Method for Low-Light Image Enhancement. *IEEE Trans. Computational Imaging*, 9:29–42, 2023. doi:10.1109/TCI.2023.3240087.

24. **S. Yang**, D. Zhou, J. Cao, Y. Guo. Rethinking Low-Light Enhancement via Transformer-GAN. *IEEE Signal Processing Letters*, 29:1082–1086, 2022. doi:10.1109/LSP.2022.3167331.
25. **S. Yang**, D. Zhou. Efficient low-light image enhancement with model parameters scaled down to 0.02M. *Int. J. Machine Learning and Cybernetics*, 15(4):1575–1589, 2024. doi:10.1007/s13042-023-01983-7.
26. **S. Yang**, D. Zhou. Single Image Low-Light Enhancement via a Dual-Path Generative Adversarial Network. *Circuits, Systems, and Signal Processing*, 2023. doi:10.1007/s00034-023-02311-8.

Teaching Preparation

Ready to teach: engineering programming; CAD/computational geometry; engineering design; intro FEA; engineering mathematics; graduate FEM; topology optimization / generative design; applied ML for design; IGA.
Relevant coursework: MECH 251/252 (FEM), MECH 415 (Optimization in Mech. Design), MECH 257 (Simulations & Modeling), MECH 259 (Design with Applied ML), MECH 294 (Design & Manufacturing), MECH 200/AMTH 201 (Adv. Eng. Math), AMTH 256 (Graph Theory), MECH 323 (Modern Control).

Skills

Python, PyTorch, MATLAB, GPU/matrix-free solvers, LaTeX; generative models; SIMP/TO; IGA/NURBS; FEA (incl. COMSOL experience); SolidWorks/CAD; Docker.

Awards & Funding Participation

China National Scholarship (Ministry of Education), 2022.
National Science Foundation (NSF) PD 22-8084 Computational and Data-Enabled Science and Engineering (CDS&E) funding opportunity through award #2245299 (research participation with Prof. Jun Wang, Santa Clara University).
NSFC Grants 62066047 & 61966037; Yunnan Key Laboratory Construction funding (with Prof. Zhou), 2020–2023.

Professional Service

Journal reviewer, *Structural and Multidisciplinary Optimization* (Springer), 2026.

Journal reviewer, *Journal of Computing and Information Science in Engineering*, 2025–2026; *Journal of Manufacturing Science and Engineering*, 2026; *Journal of Mechanical Design*, 2026; *The Imaging Science Journal*, 2025–2026; *Scientific Reports*, 2026; *Discover Internet of Things*, 2026; *npj Metamaterials*, 2026.

Conference reviewer, ASME IDETC/CIE, 2025–2026.

References

Available on request.